Project Title:

Health Insurance Cost Prediction – Machine Learning Project

This project focuses on building a complete machine learning solution to predict medical insurance charges based on demographic and lifestyle attributes. The objective is to understand how various factors such as age, BMI, smoking status, number of dependents, gender, and region influence insurance costs and to develop a reliable regression model for accurate prediction.

Project Overview

The dataset contains structured information about insured individuals, including age, gender, BMI, number of children, smoking habits, residential region, and medical charges. The project follows a complete machine learning pipeline starting from raw data preprocessing to model deployment.

Data Preprocessing & Feature Engineering

Extensive data cleaning was performed to ensure data quality and consistency. Categorical variables such as gender, smoker status, and region were encoded into numerical format for model compatibility. Feature relationships were analyzed to identify important predictors of insurance cost.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to identify trends and correlations between features and insurance charges. Visualizations were used to analyze the impact of smoking, age, BMI, and other variables on medical expenses.

Model Building & Evaluation

A Linear Regression model was trained using Scikit-learn to predict insurance charges. The dataset was split into training and testing sets to evaluate generalization performance. Model performance was measured using the R^2 score to ensure stable and reliable predictions.

Web Application Development

A Flask-based web application was developed to provide real-time insurance cost predictions. The trained model was serialized using Joblib and integrated into the backend. Users can input their details and instantly receive predicted insurance charges.

Tools & Technologies Used

Python, Pandas, NumPy, Scikit-learn, Flask, HTML, CSS, Joblib, Render (Cloud Deployment)

Key Skills Gained

Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Regression Modeling, Model Evaluation, Flask Development, Model Deployment, End-to-End Machine Learning Workflow